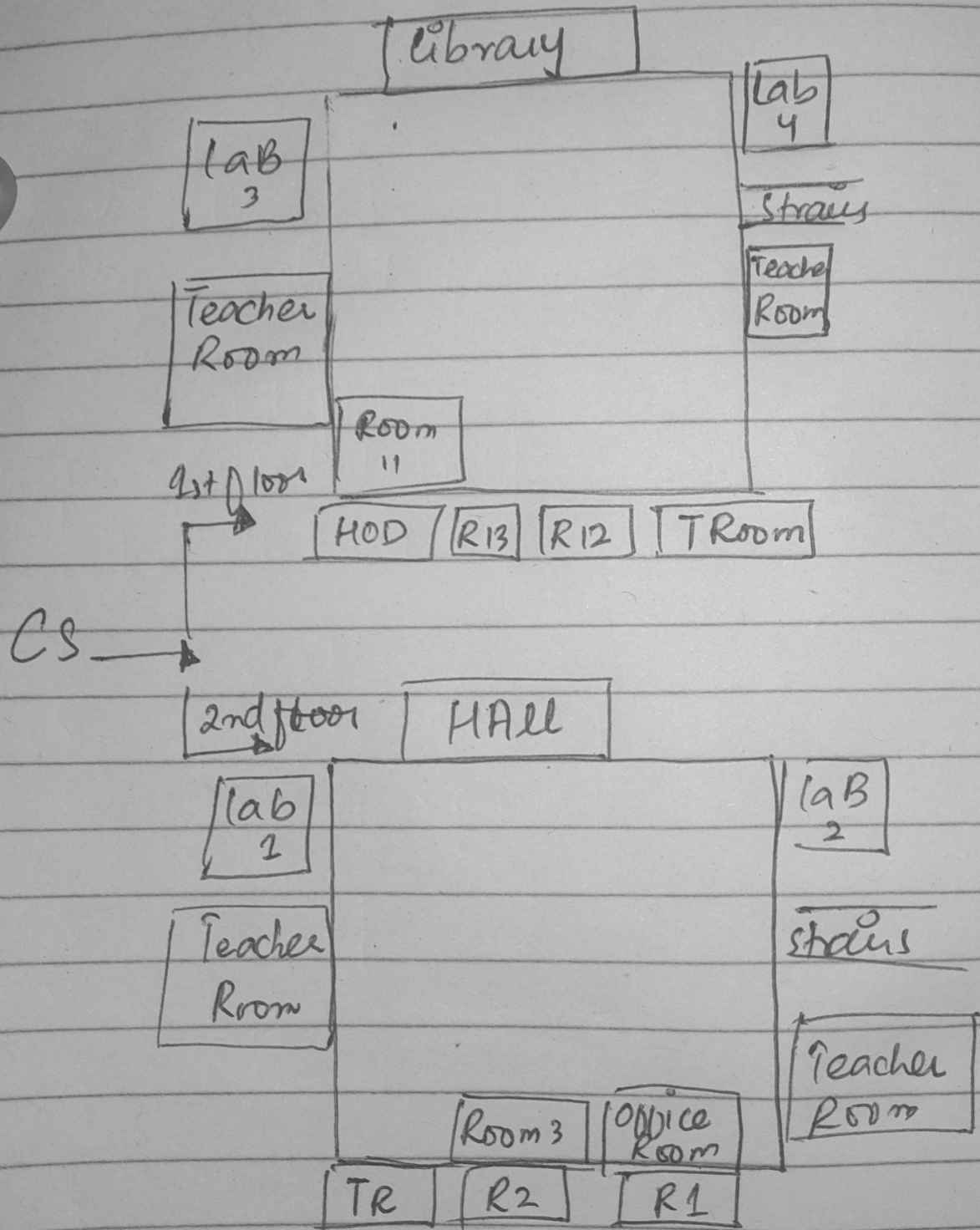
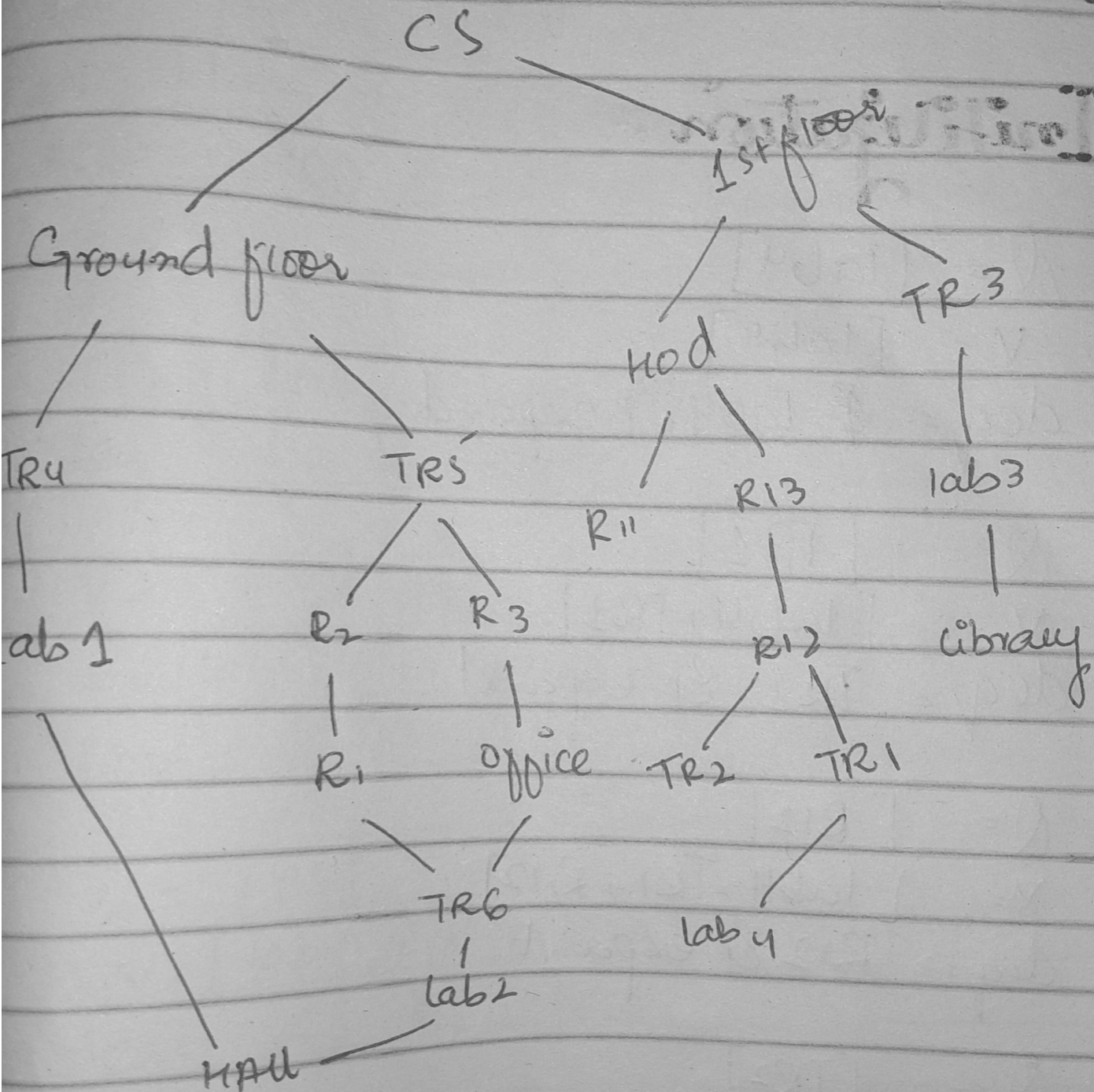


Date _____

Map of cs Department



Date _____



Q2: (lab4 to office)

Initialization :-

$$Q = [\text{lab4}]$$

$$V = [\text{lab4}]$$

$$\text{deg} = \&\text{lab4} \& \text{expand}$$

$$Q = [\text{TR1}]$$

$$V = [\text{lab4}, \text{TR1}]$$

$$\text{deg} = \text{TR1} \& \text{expand}$$

$$Q = [\text{R12}]$$

$$V = [\text{lab4}, \text{TR1}, \text{R12}]$$

$$\text{deg} = \text{R12} \& \text{expand}$$

$$Q = [\text{TR2}]$$

$$V = [\text{lab4}, \text{TR1}, \text{R12}, \text{TR2}]$$

$$\text{deg} = \text{TR2} \& \text{expand}$$

$$Q = [\text{R13}]$$

$$V = [\text{lab4}, \text{TR1}, \text{R12}, \text{TR2}, \text{R13}]$$

$$\text{deg} = \text{R13} \& \text{expand}$$

Q =

V =

deg =

[HOD]
[lab4, TR₁, R₁₂, TR₂, R₁₃, HOD]
HOD & expand

Q =

V =

deg =

[R₁₁]
[lab4, TR₁, R₁₂, TR₂, R₁₃, HOD, R₁₁]
R₁₁ & expand

Q =

V =

deg =

[1st floor]
[lab4, TR₁, R₁₂, TR₂, R₁₃, HOD, R₁₁]
1st floor & expand

Q =

V =

deg =

[CS]
[lab4, TR₁, R₁₂, TR₂, R₁₃, HOD, R₁₁, CS]
CS & expand

Q =

V =

deg =

[Ground floor]
[lab4, TR₁, R₁₂, TR₂, R₁₃, HOD, R₁₁, CS, GF]
Ground & expand

Q =

V =

deg =

[TRS]
[lab4, TR₁, R₁₂, TR₂, R₁₃, HOD, R₁₁, CS, GF, TRS]
TRS & expand

~~But that is iterative~~

$$Q = [R_3]$$

$$V = [lab_4, TR_1, R_{12}, TR_2, R_{13}, HOD, R_{11}, 1^{st} \text{ floor}, CS, GF, TR_5, R_3]$$

deg = R_3 & expand

$$Q = [office]$$

$$V = [lab_4, TR_1, R_{12}, TR_2, R_{13}, HOD, R_{11}, 1^{st} \text{ floor}, CS, Ground floor, TR_5, R_3, office]$$

deg = office & expand

Q3: Transition Table

CS $\Rightarrow$ Ground floor, 1st floor

Ground floor $\Rightarrow$ TR₄, TR₅

1st floor $\Rightarrow$ HOD, TR₃

TR₅ $\Rightarrow$ R₃, office

HOD $\Rightarrow$ R₁₁, R₁₃

R₁₃ $\Rightarrow$ R₁₂

R₁₂ $\Rightarrow$ TR₂, TR₁

TR₁ $\Rightarrow$ lab₄

Q4: Formulation

Initial state = lab 4

Goal state = office

State Space = CS department

Action = Running

Transition table = lab 4 to office

Path cost = Cost (1)

Question 5: Shortest Path

